The Pythagorean theorem and its converse

Core Concepts: The Pythagorean theorem and its converse

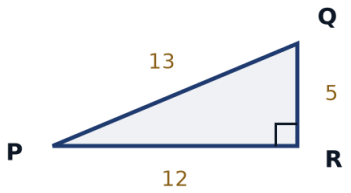
1 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 88°

b. 98°

c. 80°

d. 133°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 9

c. 7

d. 11

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 81°

b. 91°

c. 73°

d. 133°

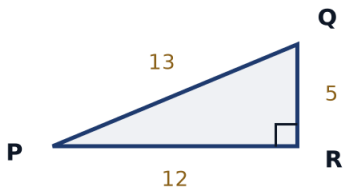
5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 7

b. 13

c. 11

d. 15

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 74°

b. 84°

c. 66°

d. 133°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

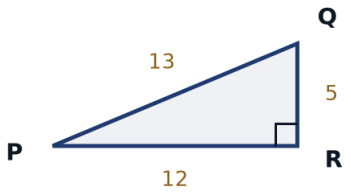
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 11

b. 17

c. 15

d. 19



Exam-Style Problem Solving & Modeling

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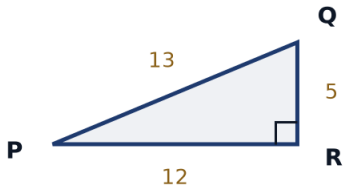
10 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 67°

b. 77°

c. 59°

d. 133°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 15

b. 21

c. 19

d. 23

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